SURAJ SINGH

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Education

Georgia Institute of Technology Atlanta, GA
Master of Science in Computer Science *Aug 2026 – Dec 2027 (expected)*

International Institute of Information Technology Bangalore (IIIT-B) Bangalore, India
Integrated B.Tech + M.Tech in Computer Science *Aug 2015 – Jul 2020*
Dean's Merit List Scholarship, ACM ICPC Chennai Regional Qualifier, Teaching Assistant (Operating Systems)

Technical Skills

Languages & Frameworks: Python, JavaScript, SQL, Bash, React, HTML, CSS, YAML, Tailwind
ML & LLM: JAX, Flax, Optax, PyTorch, LangGraph, Hugging Face Transformers, Gymnax, Brax
Infrastructure & Tools: Temporal, Docker, PostgreSQL, MongoDB, GitHub Actions, Google Cloud TPU, pytest

Experience

Gigin.ai Bangalore, India
Software Engineer III *Oct 2024 – Jun 2026*

- Designed and built a hierarchical multi-agent LLM system. Routed tasks with rule-based and LLM intent classification, RAG context, and human-in-the-loop escalation. Resulted in automation of 15% of all platform tasks.
- Architected and built the KYC verification module: document parsing, validation pipelines, and compliance workflows. Cut verification time from 1 week to under 24 hours.
- Increased release velocity and engineering throughput by 20% with CI/CD, containerized environments and deploy automation across dev and production environments.

Microsoft Hyderabad, India
Software Developer, Software Development Intern *May 2019 – Mar 2022*

- Shipped text-to-speech (TTS) in Microsoft Word for Android to 10M+ users, from intern prototype to production. Evaluated latency, voice quality, and accessibility.
- Integrated large-scale text prediction into Word mobile editors; analyzed prediction timing, latency, and UX trade-offs.
- Implemented Intelligent Reflow Mode in Word mobile: dynamic layout reconstruction, style preservation. Increased document-viewer usage by 15%.

Projects

Compute-Optimal Scaling Laws: Dense vs. MoE Transformers | *JAX, Flax, Optax* Oct 2026 – Present

- Built a transformer from scratch in JAX/Flax/Optax and a top-1, 8 expert Mixture-of-Experts variant with jax.lax.ragged_dot grouped matmuls. Trained on 10-digit addition with answer-only loss.
- Ran a Chinchilla-style IsoFLOP sweep: 106 TPU runs over 6 compute budgets (3e12 - 1e15 FLOPs) x 11 model sizes (13K - 10.6M params), dense and MoE.
- Found Chinchilla-like scaling for dense models (optimal size grows as C^0.58), but with 5 - 13x more tokens per parameter than LLMs. MoE did not outperform dense at equal compute.

Reinforcement Learning for Pac-Man | *Prof. Shrisha Rao, IIIT-B*

- Modeled Pac-Man as an MDP. Studied reward shaping, state representation, and policy convergence.
- Trained an RL algorithm agent that matched the top human score at equal play time.

Open Source Contributions

OpenEvolve | *open-source AlphaEvolve: LLM-driven evolutionary code search (Python)* Oct 2026 – Present

- Features: shipped full-rewrite mixing in diff-based evolution (v0.4.1, #580) and per-island seed programs (#547), improving best score by an average of 4% across all benchmarks.
- Fixed 4 reliability and reproducibility bugs: config not sent to worker processes (#517), runs not exiting after failed LLM retries, with tests (#545), repeated LLM seeds (#582), and lost evolution traces (#581).

Leadership & Achievements

- Top 10, Anthropic performance engineering challenge open leaderboard (868 cycles, 170x speedup).
- Professional gamer: Rank 1 NA Hearthstone Mercenaries